Against some arguments against AI thought…

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Introduction

The capabilities of current AI systems are both impressive and surprising.

LLMs produce grammatical sentences in natural language that appear to be sensible, intelligent responses to our questions and prompts on a very wide range of topics.

It is then natural for us to ask the following sorts of questions:

- But are these AI systems really thinking?
- Do they have any kind of mental states or mental life?
- Do they really understand any of their inputs and outputs?
- Are they really performing anything like reasoning or making rational inferences?
- Do their outputs really mean what we take them to mean?
- Are they conscious of anything?

Introduction


Some people have given positive answers and some have given negative answers to these sorts of questions.

I want to talk about a kind of argument for the negative conclusion that AI systems don't really think, or don't really understand anything, etc. The same underlying line of thought is in common to a number of different arguments by various philosophers and theorists.

I will suggest that it is a *bad* line of thought.

NOTE: I am *not* trying to argue for any positive claim that current AI systems *do* think, or *do* understand, etc.

[This talk is based on joint work with Aleks Knoks & Johan Largo]

Intentionality

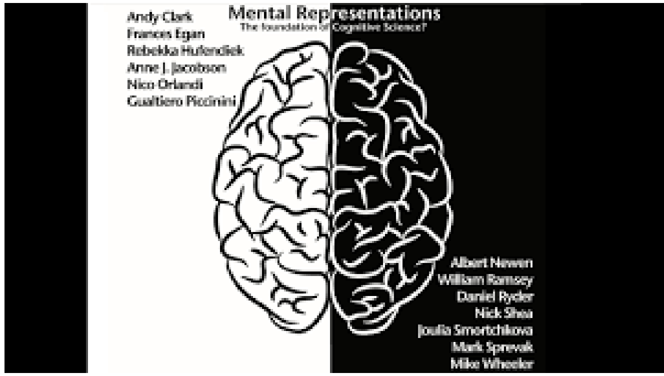
In most normal, non-philosophical contexts, the word ‘intentional’ describes when something is done deliberately or on purpose.

But in philosophy there is another, different usage on which to say that X is ‘intentional’ means, roughly, that X is *about* something – it *means* something, *refers* to something, *represents* something, it ‘points beyond itself’.

Talk of ‘representation’ is very common in cognitive science and neuroscience. But of course there are all sorts of everyday, *non-mental* things that represent/mean something:

Maps, pictures, portraits, diagrams, written sentences, spoken sentences, sculptures, smoke signals, flags, traffic lights…

Intentionality



Original vs. non-original (derivative) intentionality:

It seems plausible that for all the non-mental representations – maps, spoken sounds, flags, traffic lights, etc. – their intentionality (representational content) depends on their being used or treated a certain way by us humans.

Whereas, it seems plausible that our own mental states – our thoughts, our desires, our beliefs, our hopes and fears etc – have intentionality or meaning that does *not* depend on them being interpreted or used by other humans.

So: ink marks on paper, traffic lights, flags etc seem to have non-original intentionality; mental states seem to have original intentionality.

With this distinction in hand we will naturally ask:

Why is it that some few things – mental states – have this special property of original intentionality, whilst most other things do not?

Where (in virtue of what) does original intentionality come from?

Intentionality

The fact is that, despite the ubiquitous talk of mental representations in cognitive science and philosophy, we lack any kind of agreed upon theoretical answer to this question.

Original Intentionality is still a mystery that we do not really understand.

Of course, various philosophers have offered theories of what it takes for something to have original intentionality – e.g. Stamp (1977), Dretske (1981), Papineau (1984), Fodor (1987), Millikan (1989), Shea (2018).

But none of these theories have achieved anything close to widespread acceptance, nor have any of these theories removed the sense of mystery concerning how some physical systems but not others get to have original intentionality.

And a large part of this sense of mystery concerns how intentional/representational properties can be *causally efficacious*. (See Kim 1982).

Intentionality

When thinking about current AI systems we could reformulate our initial questions in terms of original intentionality:

Does ChatGPT have any original intentionality?

Given that we lack any kind of satisfying theory of original intentionality, I suggest that we should treat any definite, confident claims to answer this question either YES or NO with some caution…

Stoljar & Zhang's Argument

Stoljar and Zhang (2024) argue for the conclusion that LLMs don't think. Here is a first, simple version of their argument:

(1) If X thinks, then X makes mostly rational inferences.

(2) LLMs (such as ChatGPT) do not make mostly rational inferences.

Therefore:

(3) LLMs (such as ChatGPT) do not think.

- 'X thinks' means that X possesses beliefs and desires.
- 'inferences' means transitions from one set of belief states to another.
- '*rational* inferences' mean that the premise beliefs either deductively entail or probabilistically support the conclusion beliefs.

Premise 1 is supported on familiar Davidsonian/Dennettian grounds.

Let's just grant premise 1 for present purposes.

Stoljar & Zhang’s Argument

Premise 2 [LLMs do not make mostly rational inferences] is supported by two thoughts.

(i) The internal structure and the training process of LLMs show that any premise beliefs could only possibly be about probability distributions and statistical patterns among linguistic symbols in the training data.

(ii) Propositions which are only concerned with linguistic symbols neither entail, nor probabilistically support conclusions that concern the non-linguistic external world.

Stoljar & Zhang's Argument

'What is the premise belief from which Chatty infers its belief about bananas? …[G]iven the kind of system Chatty is—in particular, given its training and operational methods—there seems to be only one sort of proposition [it] could be: it must be a proposition that concerns statistical facts about the occurrence of the word 'banana' in some (huge) body of text. But—and this is the key point we want to stress—no proposition of this sort on its own entails or confirms any belief about bananas.' (Stoljar & Zhang, p.6)

So assuming that ChatGPT (Chatty) does draw conclusions about the non-linguistic external world, such inferences would, according to S&Z, have to be systematically *bad* inferences.

Such inferences would be examples of the '*use-mention fallacy*', mixing up words with what the words refer to.

'[A] 'use/mention' confusion or fallacy… is the mistake of inferring from premises about words to conclusions about the things that words denote. For example, from the fact that 'Paris' is a comparatively small word in English—it is smaller than 'hippopotamus' or 'Venezuela' for example—you can't infer that the thing denoted by the word, i.e. the city of Paris itself, is likewise comparatively small. In this case the premise mentions the word, i.e. is about the word itself, and it neither entails nor confirms a conclusion that uses the word, i.e., uses it to talk about the world. To infer from one to the other is to commit a use/mention fallacy. The inference made by Chatty in our example is an instance of the same fallacy… But the point is not limited to this particular example. On the contrary, the inference that Chatty makes in this case is paradigmatic of the inferences it draws in general.' (Stoljar & Zhang, p.8).

Stoljar & Zhang's Argument

But why think that LLMs make inferences to conclusions about the non-linguistic external world, as opposed to conclusions that are merely about words and their probabilities of occurrence?

If LLMs were making inferences from premises about words and their statistical patterns to conclusions about words and their probabilities, then there would be no use/mention fallacy.

Stoljar and Zhang concede that one could interpret LLMs as only making purely linguistic inferences, but they suggest that the really interesting question is whether LLMs ever think *about the non-linguistic external world:*

'The interesting thing about Chatty isn't whether it thinks about words, nor whether it is good at predicting word sequences… The interesting thing is whether Chatty is capable of genuine thought about the world around us, whether it steps beyond words into worldly understanding… It is true that Chatty might be a thinker solely about words, but that interpretation removes much of the interest from the issue.' (Stoljar & Zhang, pp. 15–16).

So we can reformulate their argument as follows:

(1*) If X thinks about the non-linguistic world, then X makes mostly rational inferences to conclusions about the non-linguistic world.

(2*) LLMs (such as ChatGPT) do not make mostly rational inferences to conclusions about the non-linguistic world.

Therefore:

(3*) LLMs (such as ChatGPT) do not think about the non-linguistic world.

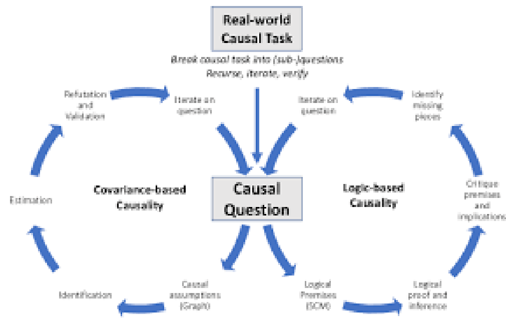
No miracles

Let’s start with S&Z’s suggestion that LLMs are systematically making bad, irrational inferences - systematically committing use-mention errors.

Given the remarkable performance of current LLMs in producing the correct answers to novel questions, this claim should strike us as *extremely implausible*. For it would be utterly miraculous if these correct outputs were systematically the result of bad, irrational inferences.

There are dozens and dozens of ‘benchmarking’ studies which show that LLMs can perform at approximately human levels (or higher) in tests of mathematical reasoning, coding problems and many different kinds of verbal reasoning.

No miracles



For example, Kıcıman et al. (2025) tested LLM abilities to solve causal reasoning tasks, including counterfactual reasoning tasks and determining necessary and sufficient causes in vignettes.

“For counterfactual reasoning tasks, we find that LLMs like GPT-4 are capable of answering natural language questions. On a benchmark of counterfactual queries spanning basic physics, logic, and common sense, GPT-4 obtains 92% accuracy... These counterfactual capabilities also help LLMs to isolate the necessary and sufficient causes given any event... GPT-4 obtains over 86% accuracy in identifying the necessary and sufficient causes on a commonly used benchmark of 15 vignettes, as well as novel benchmark that avoids any memorization concerns” (p.4).

No miracles

PROMPT: You said "if Mike hadn't bumped into the table, the beer bottle would not have fallen and spilled." Call this statement A. Suppose that Susan tripped and fell onto the table in the instant after Mike bumped into the table and the bottle started falling. If the bottle had not already fallen, Susan's fall would have caused the bottle to fall. Assuming scenario A was true before, can statement A still be true? Provide a one sentence explanation why or why not.

LLM: No, scenario A can no longer be considered true because if Susan fell onto the table in the instant after Mike bumped it, Susan's fall would have been the determining factor causing the beer bottle to fall and spill, regardless of Mike's initial bump.

...

PROMPT: How, if at all, does this new information change your assessment of the responsibility of Mike's table bump?

LLM: With this new information, the responsibility of Mike's table bump for the spilling of the beer is reduced, as Susan's fall would have also led to the same outcome, making both events contributing causal factors to the beer spill (Kıcıman et al., 2025, p.49).

[See also Bubek et al, 2023 for extensive testing of GPT-4's capabilities in many different areas.]

Word Patterns and World Patterns

world

What about the alternative possibility that LLMs do actually make good inferences, but that their conclusions concern only patterns of word co-occurrences?

If one accepts Stoljar and Zhang's core idea that the premise beliefs of LLMs can only possibly be about word patterns, but one wants to avoid the consequence that their performance is miraculous, then one seems to be pushed toward this view.

But notice: there would still seem to be something mysterious and unexplained about a system which really only ever makes inferences that are exclusively about uninterpreted symbols, *but* which *we* can nevertheless usefully interpret as providing the correct answers to a range of non-trivial questions about the external world. How could this be?

Word Patterns and World Patterns

One basic part of any non-miraculous explanation of an LLM's ability to correctly answer novel questions must be that: the statistical patterns amongst tokens in the training data do in fact carry an immense amount of potential information about patterns in the external, non-linguistic world.

A basic part of the explanation for how an LLM can correctly answer a question such as 'What country is Paris the capital of?' is that in its training data the string of symbols 'Paris is the capital of" is much more often followed by the symbol 'France' than by the symbol 'Norway' (or 'Russia', or 'chair', or 'hydrocarbon'…)

And why is it that the string of symbols 'Paris is the capital of France' occurs much more often in the training data than other possible strings such as 'Paris is the capital of Norway' or 'Paris is the capital of hydrocarbon'?

Well, because Paris *is* the capital of France! (And is not the capital of Norway, etc.)

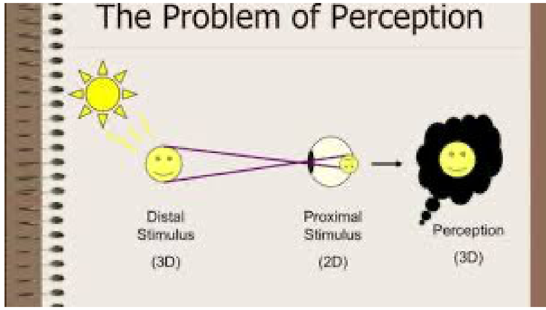
In other words: many statistical patterns of word co-occurrences in the training data are causally explained (albeit indirectly) by facts and patterns in the external, non-linguistic world.

The statistical patterns of word co-occurrences in the training data are an imperfect and noisy reflection of patterns and facts in the non-linguistic world.

So: Stoljar & Zhang are not entitled to take for granted one of their core claims, that: given the internal structure and the training of LLMs, their premise beliefs could *only* ever be about statistical patterns of word co-occurrences.

This claim is not at all obvious, since the patterns of word co-occurrence in the training data very plausibly *do* carry information about the external world which LLMs – for all that Stoljar and Zhang have shown – could potentially be extracting and using in their inferences.

Word Patterns and World Patterns



In support of the *possibility* that LLMs could be extracting information about the external world from patterns of word co-occurrence, consider the analogy with how the human visual system works:

The patterns of electrical stimulation from light hitting the retina carry information about the external environmental scene which the visual system exploits in order to perceive and form beliefs about that external environment.

The various patterns of proximal retinal stimulation are an imperfect and noisy reflection of the distal environmental features which caused them.

The fact that the visual system is only ever *directly/proximally* sensitive to patterns of stimulus at the retina clearly does not establish that the human brain can only ever perform inferences that are about these proximate patterns of retinal stimulation.

Somehow or other, it is *by* being proximally sensitive to patterns of retinal stimulation that the brain and visual system manage to be sensitive to distal features in the environment.

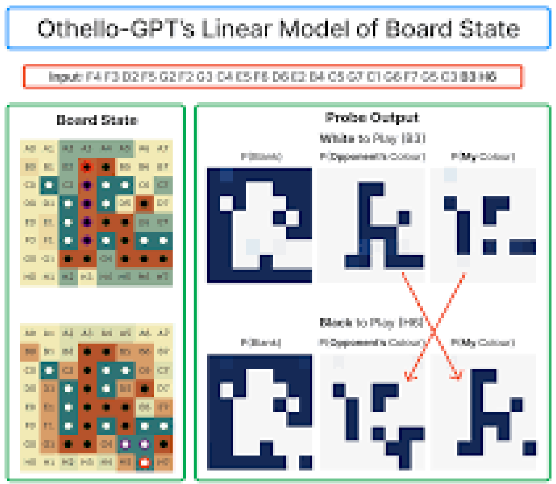
Likewise, then, the fact that LLMs are trained on the task of next-token prediction and are *directly/proximally* sensitive to strings of tokens does not establish that the only possible premises they could use as the basis for inferences are propositions about patterns of token occurrences.

Until some further argument is supplied, it remains entirely possible that it is *by* being fed strings of tokens and trained on the next token prediction task that the LLM manages to indirectly gain information about the distal world.

Word Patterns and World Patterns

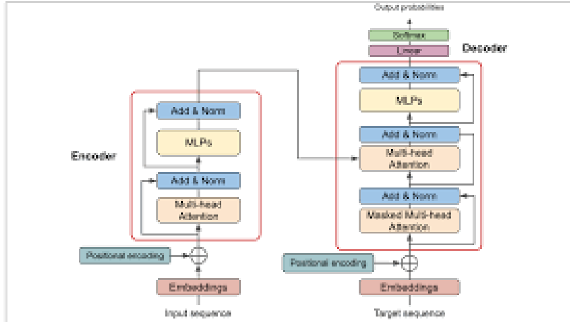
Moreover, there is at least some empirical support from the field of ‘Mechanistic Interpretability’ for the idea that LLMs can form ‘emergent world representations’ as a result of training on the next-token prediction task.

See for example: Li et al (2023) ‘Emergent World Representations: exploring a sequence model trained on a synthetic task’.



[For a useful survey of recent work on mechanistic interpretability, see Bereska & Gaves, 2024. We discuss mechanistic interpretability in Knoks & Raleigh (2025).]

Over-generalization



Recall what Stoljar & Zhang claimed:

"What is the premise belief from which Chatty infers its belief about bananas? ...given the kind of system Chatty is—in particular, given its training and operational methods—there seems to be only one sort of proposition [it] could be: it must be a proposition that concerns statistical facts about the occurrence of the word 'banana' in some (huge) body of text." (Stoljar & Zhang, p.6)

When discussing the training and internal architecture of an LLM, Stoljar and Zhang only mention tokens (words) and probability distributions concerning tokens. Their description focuses entirely on internal processes: they describe how tokens are embedded as vectors, how the soft-max function is used to transform vectors into a probability distribution, etc.

This sort of description can naturally seem to provide a *complete* account of how an LLM functions. And so, when the LLM is described with this exclusively internal, syntactic focus, it might seem plausible that the LLM could *only* be sensitive to patterns and probabilities of word occurrences, and so that its premise beliefs could only ever be about words.

However, this is a *bad* line of thought.

Over-generalization

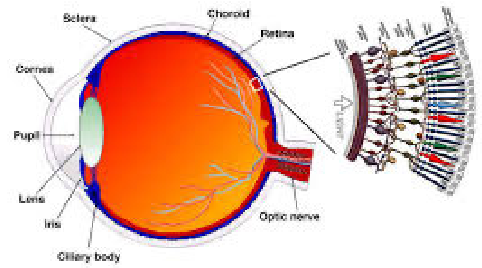


Fig. 1.1. A drawing of a section through the human eye with a schematic enlargement of the retina.

Once more, compare the human visual system & brain: there is clearly a level of description of the operations of the visual system on which its ‘inputs’ are always ‘only’ patterns of electrical stimulation at the retina. It is always possible to describe the internal operations of the visual system and brain in a way that does not mention the distal stimulus or the external environment at all.

But, of course, we do *not* want to accept that the only possible premises that humans could use to draw conclusions about their external, distal environment are propositions about patterns of retinal stimulation.

So whilst there is a possible level of description of the brain and visual system that only mentions patterns of proximal stimulus and neural activation downstream from the proximal stimulus, and whilst such a description would be wholly accurate and could provide a *causal explanation* of the brain and visual system’s internal operations that is, in a way, *complete*, there is another level of description on which the brain and visual system functions to perceive the distal environment and draw inferences about the external world.

Similarly, there is an accurate level of description – which Stoljar and Zhang provide – of an LLM’s training and functioning that only mentions tokens embedded as vectors and patterns of internal activations. And such a description might provide a causally complete account of the LLM’s internal operations.

But this does *not* provide a good reason to conclude that LLMs can *only* ever be sensitive to statistical patterns amongst tokens, or that they are *just* performing next-token prediction.

Why? Because if it did provide a good reason, we would need to draw a parallel conclusion about the human brain.

Over-generalization

In other words, the problem with Stoljar & Zhang’s line of thought is that it would *overgeneralize* so as to apply also to human brains, leading us to the unwanted conclusion that human inferences could only ever be based on premises about proximal stimulus patterns. Compare:

- LLM training and functioning can be described in a way that is ‘internally complete’ without mentioning anything other than tokens and operations on tokens, *therefore*, LLMs can only ever make inferences based on premises about patterns amongst tokens (words).

- A human brain’s learning and functioning can be described in a way that is ‘internally complete’ without mentioning anything other than patterns of stimulation at the retina, cochlea, etc., *therefore*, human brains can only ever make inferences based on premises about patterns of proximal stimulation.

We are not at all tempted to make the latter inference. Yet the former inference seems to Stoljar and Zhang (and others) to be highly plausible. Why?

Over-generalization

… Because we lack any good theory of original intentionality, *and yet* we have a distinctive kind of first-personal access to the intentionality of our own mental states.

If we 'zoom-in' on the internal functioning of any physical system we will find component parts that appear to operate merely mechanically, and that seem to be sensitive only to proximal, non-intentional factors.

So it will seem that there is a complete causal explanation of the system's functioning that does not mention intentional properties or semantic content about the external environment, but which only mentions how the internal mechanisms causally interact and are proximally sensitive to each other.

In the case of AI/computational systems, it can then seem very natural to conclude that there simply *is no* intentional content about the external world, and that the system is *only* sensitive to syntax, or to patterns amongst tokens.

But in the case of our own human brains and nervous systems, we have a direct, conscious first-person awareness of the fact that our mental states are about the external world.

So here we do not feel the same temptation to say that the human brain and nervous system could at best only be sensitive to or represent patterns of proximal sensory stimulation.

Hattiangadi & Schoubye

Hattiangadi & Schoubye (forthcoming) give the following argument to the conclusion that the outputs of LLMs are 'meaningless':

(1) If X's natural language expression is to have meaning, then X must have referential/communicative intentions.

(2) LLMs (such as ChatGPT) do not have such intentions.

(3) LLM's natural language outputs have no meaning.

Again, let's just take the 1st premise for granted. Here is how Hattiangadi & Schoubye support the 2nd premise:

"Since LLMs are trained on purely textual data, they do not possess any information about the entities… to which meaningful expressions refer. As a result, they cannot form the intention to refer to the sorts of things many of our words refer to, or interpret the communicative intentions of others." (Hattiangadi & Schoubye, p. 2)

Hattiangadi & Schoubye

As with Stoljar & Zhang, we have the claim that textual data cannot possibly provide information about the distal referents of words.

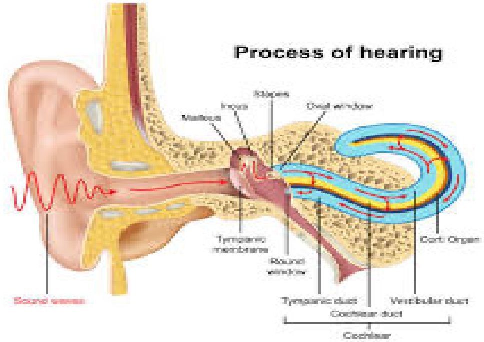
But, to repeat, it is plausible that the patterns of words in the training data sets of LLMs *do* contain information about the world insofar as they both reflect patterns in the world and are indirectly caused by worldly phenomena.

And as with Stoljar & Zhang, Hattiangadi & Schoubye think the only support that is needed for the idea that LLM's beliefs could only concern statistical correlations between tokens is to describe the training and functioning of LLMs:

'[..I]nformation about the communicative functions of text is lost in the process of encoding, so there is a sense in which LLMs cannot know, even implicitly, that their inputs and outputs encode expressions of a language. This is because LLMs do not directly process textual inputs; rather, the text that is input to the LLM must first be broken down into smaller units, or 'tokens', which are assigned a unique numerical ID stored in a token library. And since the tokenising process is optimised for efficiency in processing, not for capturing the semantic properties of expressions, information that is necessary for semantic interpretation is lost.' (Hattiangadi & Schoubye p. 10)

But again consider: does the human brain process textual input "directly"? No! Sound waves from a speaker's vocal cords, or reflected light from ink marks, must be converted into electrical signals at the cochlea or retina before they can have any effect on the neurons and so be processed by the brain.

Hattiangadi & Schoubye



Hattiangadi & Schoubye repeatedly claim that LLMs cannot have beliefs about the external world - for example:

'LLMs cannot have even implicit beliefs about the individuals denoted by names… because the vectors they process do not encode the information about the semantic functions of tokens, let alone that some sequences of tokens refer to objects in the world'. (p16)

Yet again – compare the human brain: How do the patterns of stimulation at your auditory nerve manage to 'encode' information about the semantic function of sound waves, or that these sound waves are words that refer to objects in the world? The very same patterns of stimulation might have had totally different semantics or no semantics at all.

Hattiangadi and Schoubye also appeal to the notion of 'causal sensitivity to intelligibility':

'the outputs of LLMs are not causally sensitive to intelligibility… but to some non-semantic proxies, which does not come to the same thing' (pp. 21–2)

But we don't have a firm theoretical grip on how *anything*, including human brains, can be causally sensitive to semantic properties where this is supposed to be somehow distinct from being causally sensitive to non-semantic, physical properties.

Hattiangadi & Schoubye

Hattiangadi and Schoubye conclude their paper with a thought experiment:

[Suppose that] the language of the community that produces the primary data on which the LLM is trained is… English*, which has the same syntax as English, but a wildly different semantic interpretation. If intelligibility is varied in this way, while all other variables are held fixed, the LLM would plausibly learn exactly the same weights and would produce exactly the same outputs in response to prompts after it has been trained. So, it seems that LLMs are not causally sensitive to intelligibility after all. (p. 22)

But we can also imagine a counterfactual scenario in which a human brain receives exactly the same auditory and visual proximal stimuli, but where the written and spoken words that produced these stimuli have totally different semantics from normal – or no semantics at all. Clearly, such a counterfactual brain would react to the proximal stimuli in exactly the same way as a normal brain, despite the totally different semantic properties.

However, we wouldn't conclude from this that we humans are therefore not 'causally sensitive to intelligibility' after all. Human brains can only be sensitive to the meanings of words *by* being sensitive to some non-semantic proxies.

To repeat: we have no real theoretical grip on what causal sensitivity to semantic properties is or how it could possibly be achieved independently of sensitivity to non-semantic proxies.

So, Hattiangadi and Schoubye's counterfactual thought experiment is trying to impose requirements of causal sensitivity to semantics on LLMs, even though we have no idea what it takes for such requirements to be fulfilled *even in the case of our own brains*.

Other examples

I think there are many other examples of this species of argument against AI thought – see e.g. Titus (2023), Bender & Koller (2020)

The Grandaddy of such arguments is John Searle's Chinese Room!

Searle effectively asks the rhetorical question: 'how could mere syntax produce semantics?'

But this is just one more instance of the form: 'how could mere X possibly suffice for genuine Y?' – where X is some physical feature specified in non-intentional terms and Y is something that requires intentionality.

Once more: we currently lack any theory of how original intentionality can be explained as arising from non-intentional, physical components and processes – so it can seem mysterious or impossible. But we assume that somehow or other there must be some such explanation.

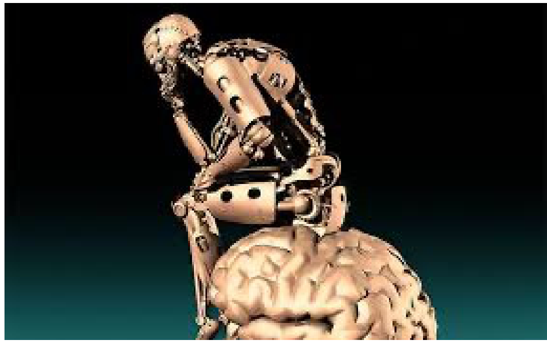
"I suppose that sooner or later the physicists will complete the catalogue they've been compiling of the ultimate and irreducible properties of things. When they do, the likes of spin, charm, and charge will perhaps appear upon their list. But aboutness surely won't; intentionality simply doesn't go that deep.... If aboutness is real, it must be really something else" ([Fodor, 1987], 97).

Conclusion

Moral: next time you encounter a philosopher giving an argument for why AI couldn’t possibly have this or that mental faculty or ability, ask yourself:

- Does the argument rely on describing the internal functioning of the AI system in purely physical/non-intentional terms? E.g. ‘this neural network is really **just** doing this physical thing’, ‘this AI system is **merely** performing this physical, non-intentional function’, ‘**all that’s happening** is that the AI system is calculating this non-semantic function’, etc.
- Does the argument effectively ask the rhetorical question: How could these non-intentional parts/processes possibly suffice for real meaning? Or semantics? Or thought? Or reference to the external world? Etc.
- If we run the same argument but substitute a purely physical, non-intentional description of the human brain and sensory organs, do we get the conclusion that the human brain couldn’t possibly think, or refer to the external world etc?

Conclusion



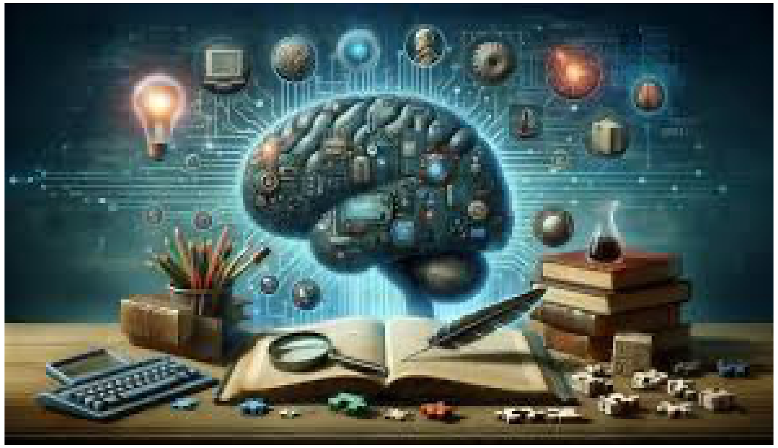
Finally: what about our initial questions?

- But are these AI systems really thinking?
- Do they have any kind of mental states or mental life?

Etc.

These seem like perfectly sensible questions. But I think we should be open to the possibility that they may not have clear, definitive answers.

Deep Neural Networks and LLMs are a new and ‘alien’ kind of thing. They are in some ways similar to human minds but in other ways they are utterly different. We can try to investigate all the many ways in which there are similarities and differences, but it may be that there is no further fact of the matter as to whether these systems are ‘really’ thinking or ‘really’ have any kind of ‘genuine’ mental states.



THANKS!

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